

PKCERT AI & Software Development Internship

Task 05 – Data Preprocessing, Visualization & SQL Fundamentals

Objective:

Develop practical skills in data preprocessing, feature engineering, data visualization, and SQL-based data extraction using real-world datasets.

Total Marks: 100

Part A – Data Cleaning & Preprocessing (30 Marks)

1. Select a public dataset from Kaggle or another open data source.
2. Handle missing values, duplicate records, and outliers.
3. Prepare a brief report explaining each preprocessing step.

Part B – Data Visualization (25 Marks)

1. Create a Histogram, Boxplot, Correlation Heatmap, and two additional visualizations using Matplotlib and Seaborn.
2. Explain the insights obtained from each visualization.

Part C – Feature Engineering (20 Marks)

1. Identify categorical features.
2. Apply One-Hot Encoding and/or Label Encoding.
3. Perform normalization or standardization and explain its importance.

Part D – SQL Fundamentals with Python (25 Marks)

1. Create a small relational database from the cleaned dataset.
2. Perform SELECT, WHERE, GROUP BY, and JOIN queries.
3. Connect Python to the database and display query results.

Assessment Area	Marks
Data Cleaning & Preprocessing	30
Data Visualization	25
Feature Engineering	20
SQL & Python Integration	25

